# Anh Quang Nguyen

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I'm an Anthropic Claude Certified Architect and a part-time Stanford student taking accelerator physics and quantum computing in Fall 2026. I care about safety, security, alignment, and well-being at scale. I'm seeking opportunities starting in Winter/Spring 2027 with a happy mission-driven team. I enjoy problem-solving and working through math at a whiteboard, and I'm eager to learn quickly and contribute.

## PROJECTS & EXPERIENCE

**GPU acceleration with CuPy**

Python / CuPy / PyTorch

- Wrote **CuPy** (Python) for GPU-accelerated pair-counting on large datasets.
- **AI field-level inference**: GPU forward-modeling with neural-posterior training (SBI).

**IC design: RTL-to-GDSII flow for extreme / high-radiation environments** *TSMC / ASAP7 PDK*

Cadence / Synopsys

- Drove the complete design flow to DRC/LVS-clean, signoff-complete GDS, radiation-hardened to roughly **300 krad(Si)** total ionizing dose (TID).

**AI-powered mobile device management (MDM): safety, security, privacy & digital wellbeing**

iOS · MDM · AI · security / privacy *2024-2025*

- Built a privacy-first, AI-powered MDM platform as an alternative to Jamf School and Jamf corporate, applying AI across device safety and security: policy enforcement, secure configuration, and privacy controls, with age-appropriate child safety as one application.
- Shipped device-level usage and content controls in **2024-2025**, ahead of the industry: Apple has since built equivalents into iOS 27, and Meta agreed to default teen time limits, night mode, and school-hours muting in its Aug 2026 settlement with state attorneys general.

**Full-stack Web3 engineering & on-chain trading** *Team lead*

JavaScript / React · AWS *2021-2024*

- **Led a small team** to ship a production Ethereum application end-to-end (a **JavaScript / React** frontend and an **AWS** backend), reaching roughly **$3M in annual recurring revenue (ARR)**.
- Engineered the AWS backend to sustain heavy concurrent traffic, **stress-tested to hold up under high web load** at launch.

## CONTRIBUTIONS

**SLAC National Accelerator Laboratory** *Menlo Park, CA*

Community white papers & workshops *2026*

- Two 2026 community white papers: **Rubin 2036** (KIPAC workshop, Aug 2026), and a **Dedicated AI/ML Accelerator Test Facility** paper in progress, following the Jul 2026 SLAC workshop on ML-driven accelerator controls and diagnostics.

**Upstream contributions**

NVIDIA NemoClaw · OpenShell · NeMo Curator / Retriever · NASA F Prime

- Contributing to migrating the architecture to run on **OpenShell**: the sandboxed, policy-governed agent runtime behind **NemoClaw**, used across NVIDIA teams.
- Security test merged into **NemoClaw**, a static-analysis fix merged into **NASA F Prime**, and a checksum-verification gap in **OpenShell** that NVIDIA fixed after I reported it.

## TECHNICAL SKILLS

**GPU & TPU stack:** NVIDIA GPU (CUDA, GH200, A100, T4, CuPy, PyTorch) and Google **Cloud TPU** (v5e, v6e, v5p) across the full **XLA** stack (JAX, Flax, sharding meshes, HLO-level lowering, step-time profiling), on **GKE** clusters (Autopilot, GPU/TPU node pools) and SLURM, with Docker/Apptainer image pinning and multi-zone TPU quota and capacity management; benchmarked one training workload across five architectures and three sites.

**Scaling & MoE architecture:** measured **all-to-all** (the MoE routing primitive) against all-reduce from 8 to **256 chips** on a multi-host TPU v6e slice, isolating the routing tax that grows with expert parallelism; **SparseCore** gather behaviour on v5p/v6e, HBM-promotion thresholds, and torus-shape and interference effects on collective tail latency.

**Fine-tuning: LoRA/PEFT** and teacher-model distillation for open frontier-class models (NVIDIA Nemotron, Qwen3) on 8x A100 via Hugging Face Jobs and Modal, with merged-safetensors and GGUF quantization export.

**ML / GPU computing:** Python, autodiff; GPU quantum optimal control, ML calibration & readout, device inverse design, ML spectral (ARPES / pump-probe) analysis.

**Multiphysics simulation:** COMSOL (EM/RF, thermal, structural); microwave resonators & cavities, optomechanical / photonic devices, MIM / THz instrument modeling.

**Instrumentation & control:** FPGA qubit-control stacks (QICK), DAQ automation, measurement orchestration, reproducible data pipelines, full-stack dashboards.

**Safe automation:** sandboxed, policy-governed agent runtimes (OpenShell-style) for human-in-the-loop closed-loop experimentation.

**Semiconductor / device:** full RTL-to-GDSII on TSMC / ASAP7 7nm in Cadence/Synopsys.

**Tooling:** Python (NumPy/SciPy/pandas), C/C++, shell, Git, Docker.

## EDUCATION

**Stanford University** *Stanford, CA*

Visiting Researcher *2026*

Graduate coursework, School of Engineering and Graduate School of Business (part-time) *Cumulative GPA: 4.3*

- Presented *Train, Tune & Scale Qwen3 on Cloud TPUs* at **Google**: TPU v5e/v6e/v5p benchmarking, **97% strong-scaling efficiency** on 16 v6e chips.
- Found that MFU tracks a chip's **bytes-per-FLOP ratio** rather than its peak FLOPs.

**University of California, Los Angeles (UCLA)** *Los Angeles, CA*

International Institute *2019*

## SAFETY TRAINING & CERTIFICATIONS

**Stanford SNF / EH&S:** EHS-4200 General Safety · EHS-1900 Chemical · EHS-2200 Compressed Gas · EHS-4300 Hazard Comm · CHEM-1000.

**Certifications:** Anthropic **Claude Certified Architect, Professional**, 2026; NVIDIA AI certificates in progress.